

DIGITAL ELECTRONICS AND PULSE TECHNIQUE

LAB WORK

CSE 2366

Fall 2025

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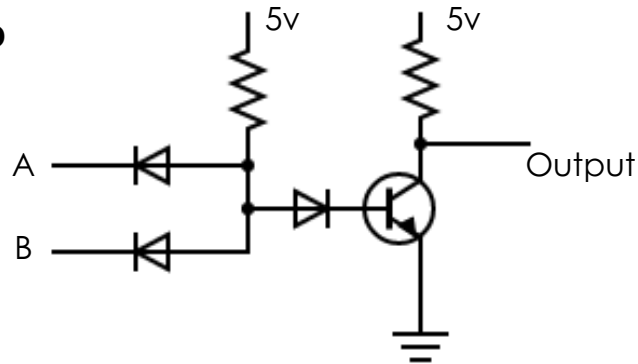


Experiment No. **04**

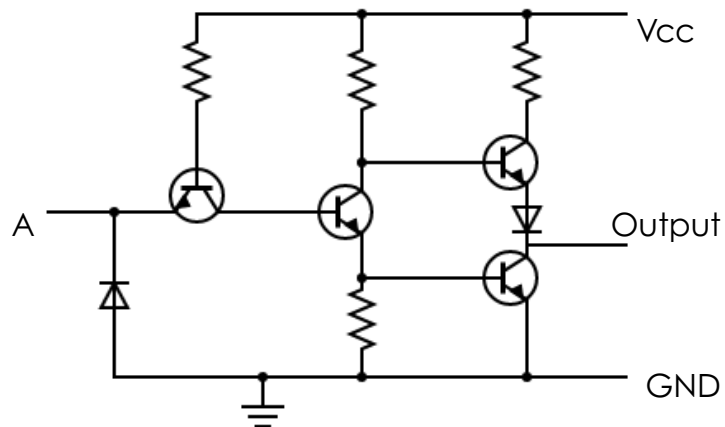
Experiment Name : **DTL- NAND, TTL - INVERTER & NOR**

4.1 Circuit diagram

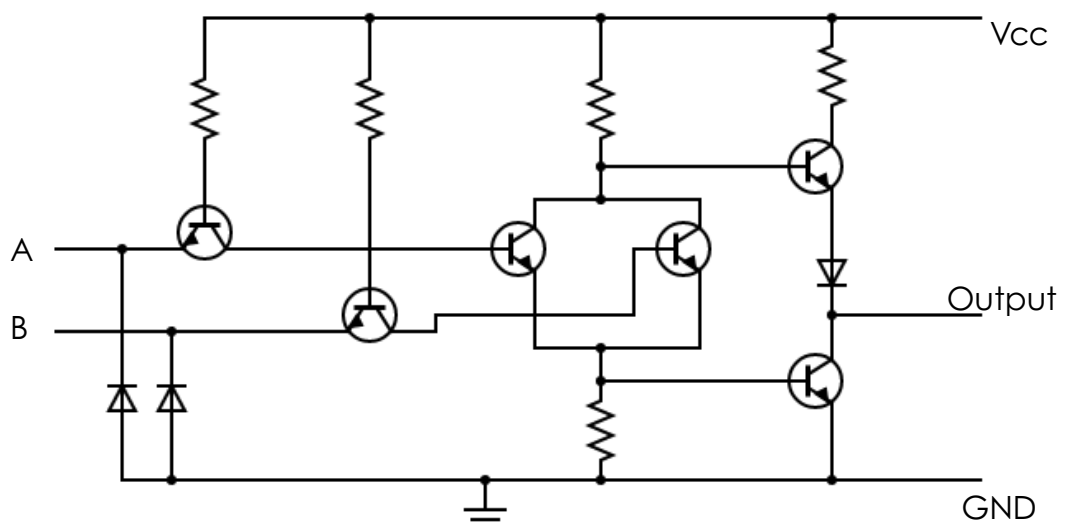
DTL-NAND



TTL-INVERTER

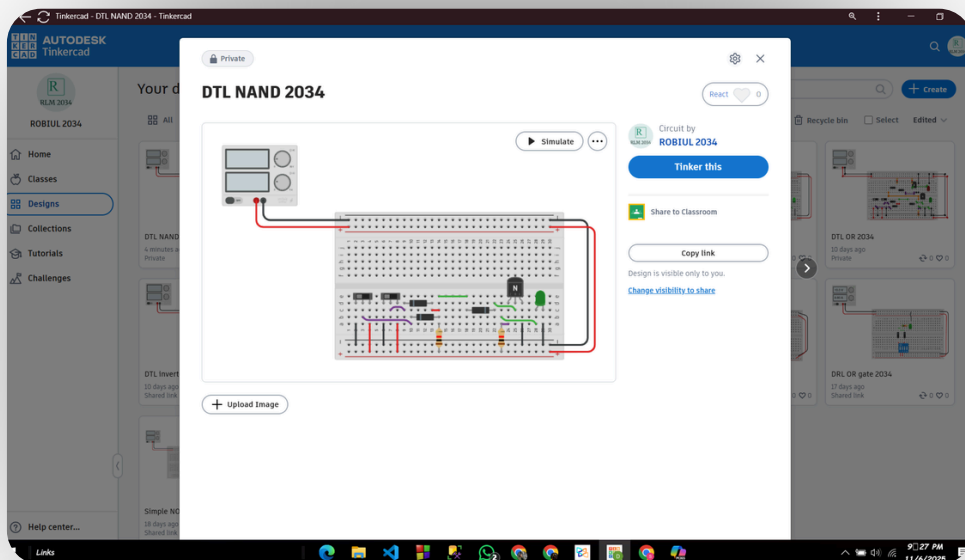


TTL-NOR

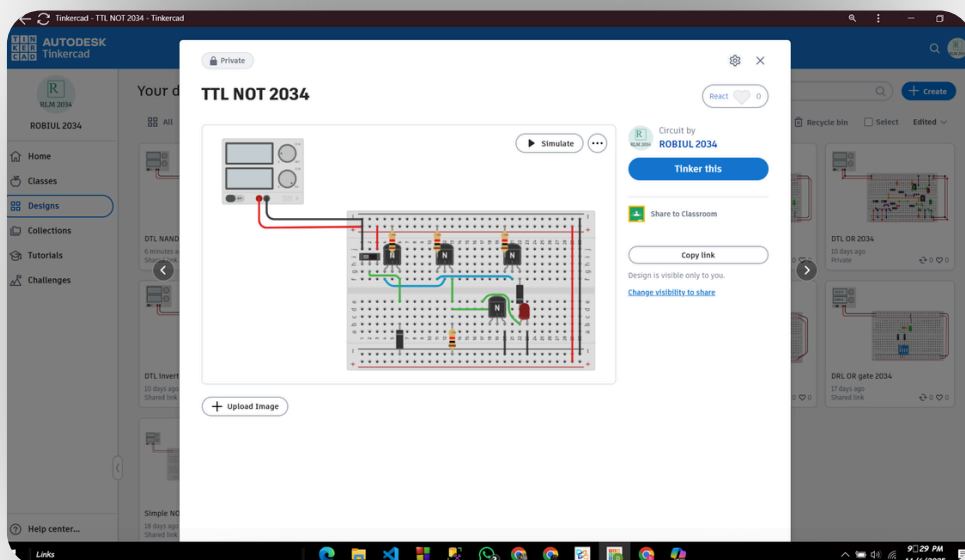


4.2 Circuit screenshots

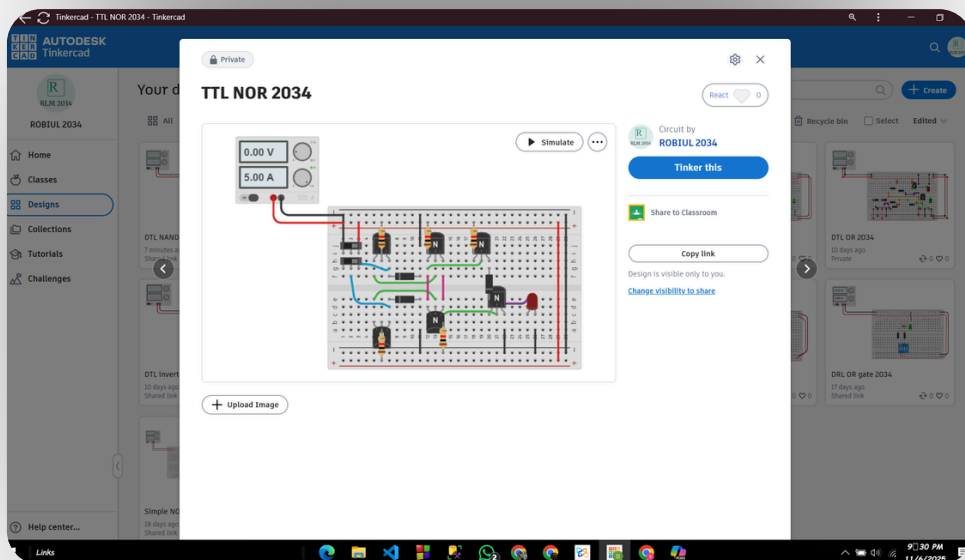
DTL-NAND



TTL-INVERTER



TTL-NOR



4.3 Circuit links

DTL-NAND [Click Here](#)

TTL-INVERTER: [Click Here](#)

TTL-NOR: [Click Here](#)

4.4 Discussion

DTL NAND Gate:

In the DTL NAND circuit, diodes and a transistor are used to perform the NAND logic operation. The gate produces a low output only when all inputs are high. If any input is low, the output remains high. This happens because the diodes prevent the transistor from turning on unless all inputs are high. The experimental observation matched the NAND truth table, confirming correct operation.

TTL Inverter:

The TTL inverter uses a transistor circuit to invert the input signal. When the input is high, the transistor conducts and the output becomes low. When the input is low, the transistor is off, and the output becomes high. This shows that the output is always the opposite of the input. The observed results verified the inverter action properly.

TTL NOR Gate:

In the TTL NOR gate, multiple transistors are used so that the circuit gives a high output only when all inputs are low. If any input is high, one of the transistors conducts, pulling the output low. This behavior follows the NOR logic rule. The experimental result was consistent with the NOR gate truth table, confirming the expected logical performance.